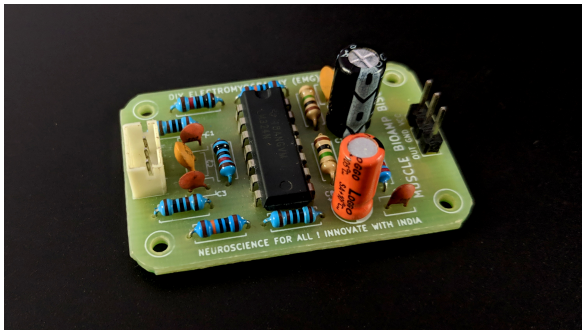
Muscle BioAmp Biscute

Affordable DIY ElectroMyography (EMG) sensor for Human-Computer Interface projects

Overview

Muscle BioAmp BisCute is the most affordable DIY ElectroMyography (EMG) sensor in the world that allows you to create a Human-Computer Interface (HCI) projects with ease. In the process of building your own BisCute, you learn what goes into making a functional biopotential amplifier that can be used for amplifying sub mV signals generated by muscles inside your body to a level a microcontroller unit (MCU) can understand.



Specifications

Minimum Input Voltage	3.3-30 V
Input Impedance	10^{11} ohm
Fixed Gain	x2420
Bandpass Filter	72 – 720 Hz
Compatible Hardware	Any development board with an ADC (Arduino UNO & Nano, Adafruit QtPy, STM32 Blue Pill, BeagleBone Black, Raspberry Pi Pico, to name just a few) or any standalone ADC of your choice
BioPotentials	EMG (Electromyography)
No. of Channels	1
Electrodes	3 (Positive, Negative, and Reference)
Dimensions	3.0 x 4.5 cm
Open Source	Hardware + Software

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1 Muscle BioAmp Biscute

1.3 Hardware

Images below shows a quick overview of the hardware design.

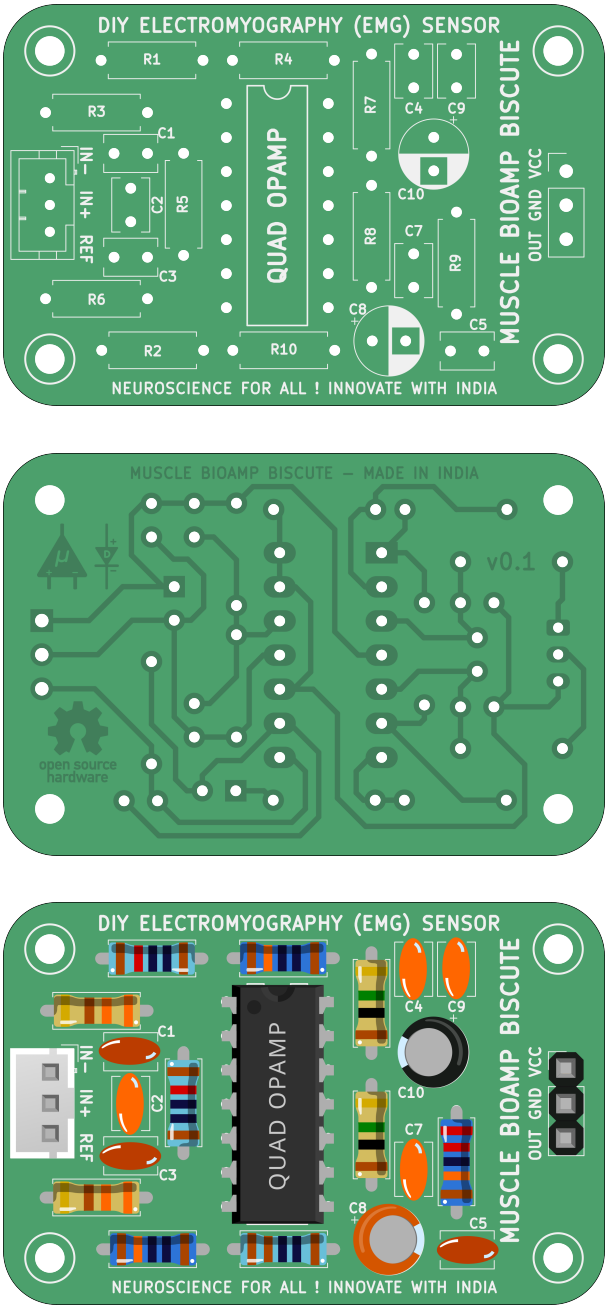


Fig. 1: Assembled PCB

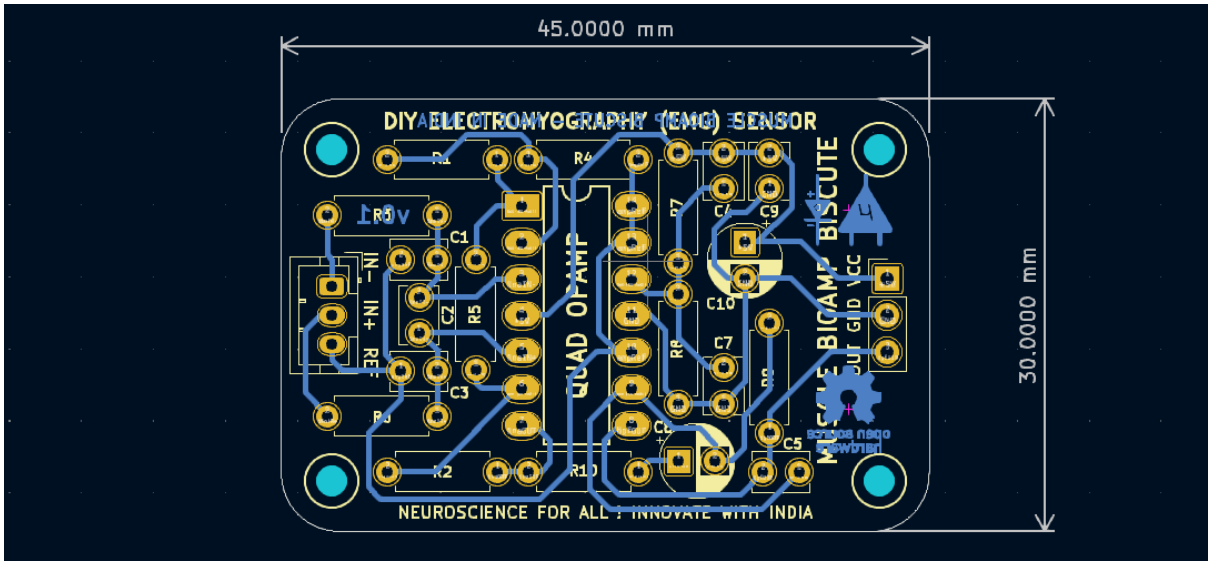


Fig. 2: PCB Layout

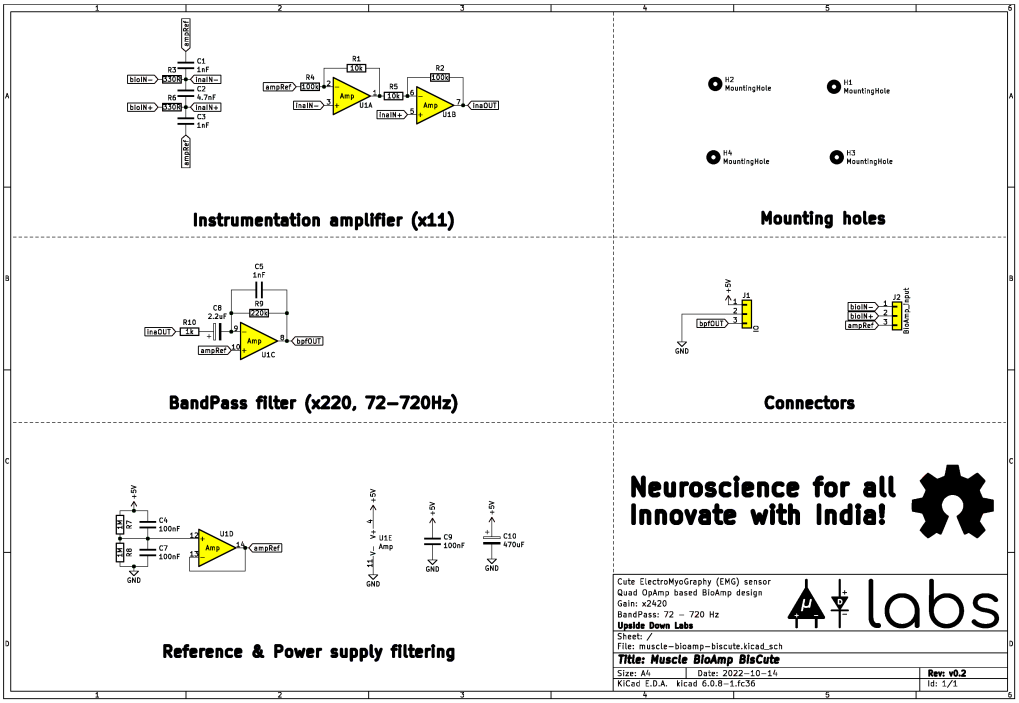
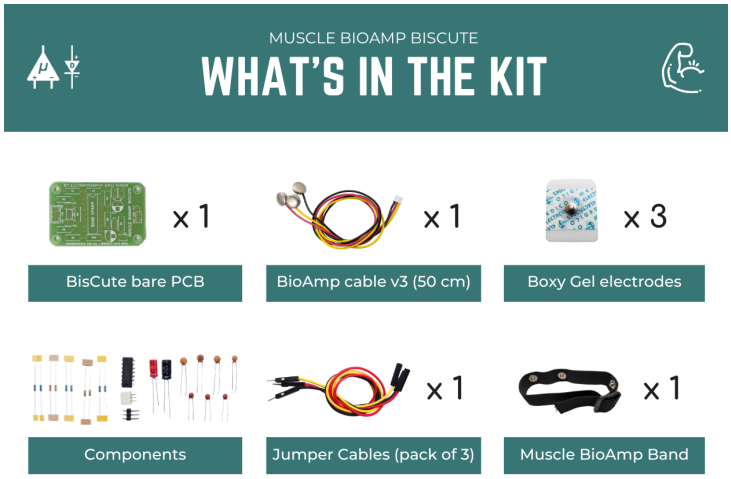


Fig. 3: Schematic Diagram

1.4 Contents of the kit



1.5 Software requirements

- Before you start using the kit, please download Arduino IDE v1.8.19 (legacy IDE). Using this you'll be able to upload the arduino sketches on your development board and visualise the data on your laptop.

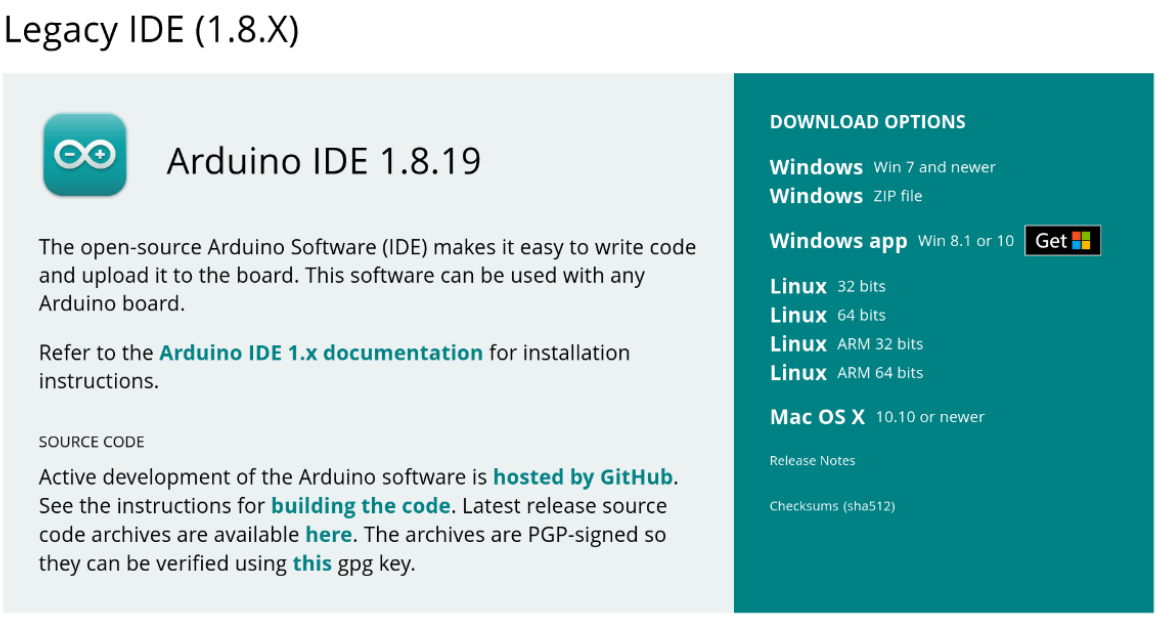


Fig. 4: Arduino IDE v1.8.19 (legacy IDE)

Visit Upside Down Labs Chords Web to visualize your bio-potential signals directly in the browser. Flash the Chords-Arduino-Firmware on your board to visualize signals on Chords Web.

Getting started with Chords Web

https://youtu.be/IVIPnk9z75g

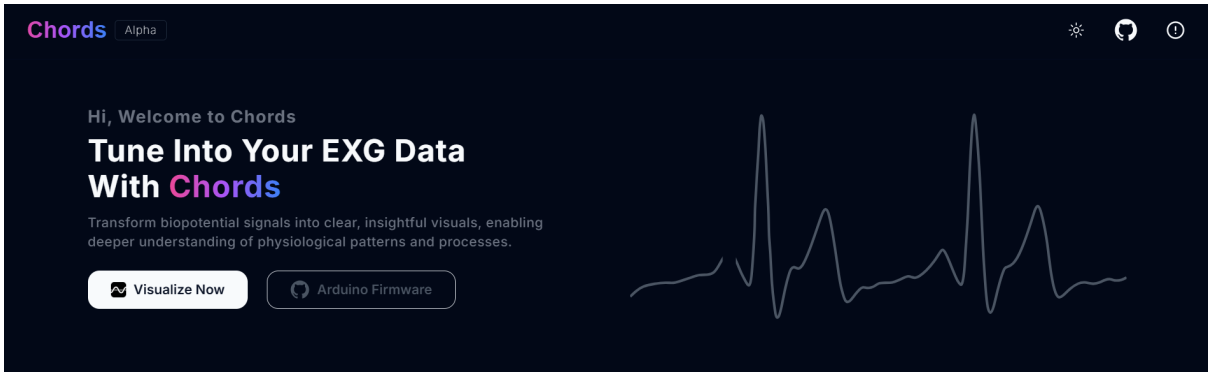


Fig. 5: Chords Web Landing Page

1.6 Assembling the kit

You can get Muscle BioAmp BisCute from our online stores (shipping worldwide) and for assembling the BisCute you can either take a look at this interactive BOM or the step by step guide given below.

Note: Follow the highlighted yellow shapes to assemble your Muscle BioAmp BisCute!

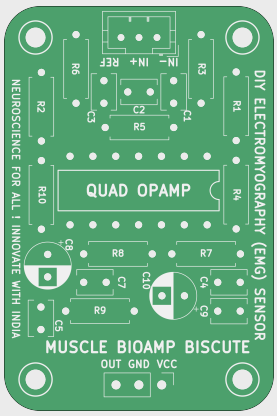


Fig. 6: Bare PCB

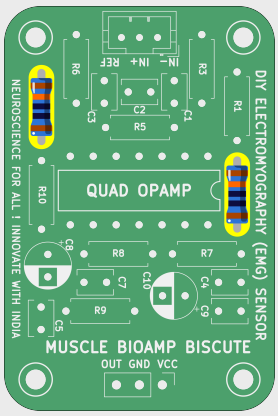


Fig. 7: 100K Resistor

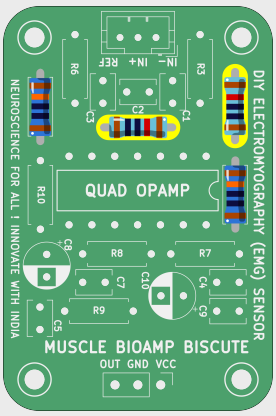


Fig. 8: 10K Resistors

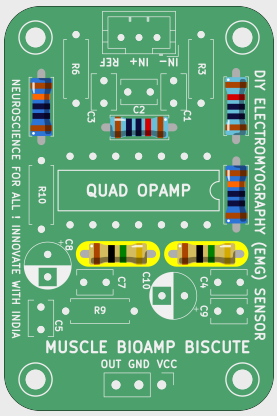


Fig. 9: 1M Resistors

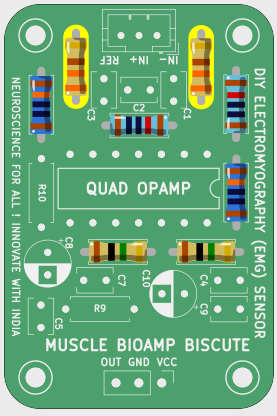


Fig. 10: 330R Resistors

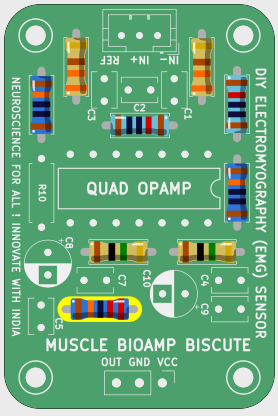


Fig. 11: 220K Resistor

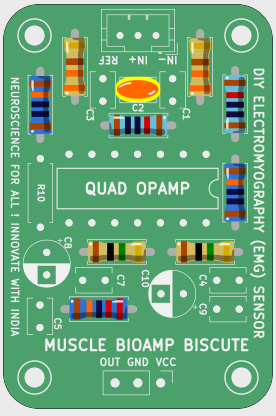


Fig. 12: 4.7nF Capacitor

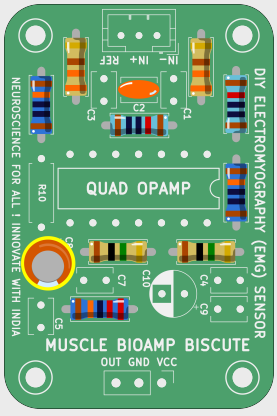


Fig. 13: 2.2uF Capacitor

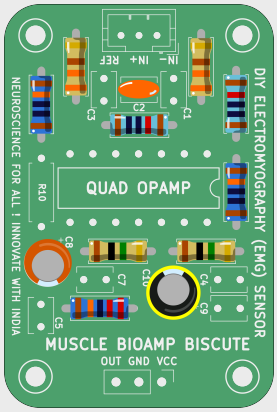


Fig. 14: 470uF Capacitor

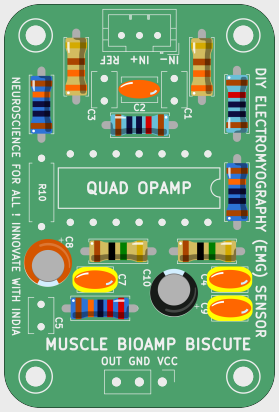


Fig. 15: 100nF Capacitors

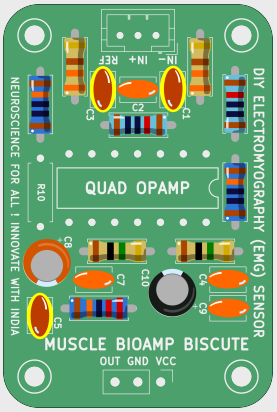


Fig. 16: 1nF Capacitors

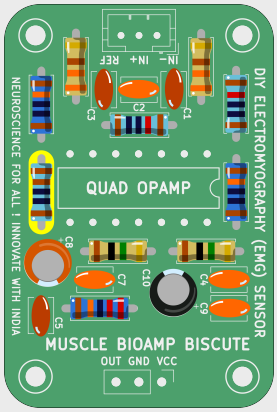


Fig. 17: 1K Resistor

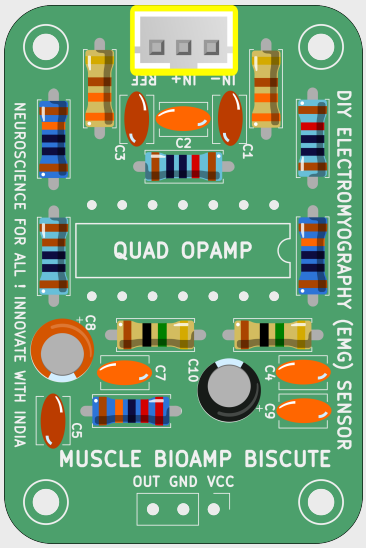


Fig. 18: BioAmp Connector

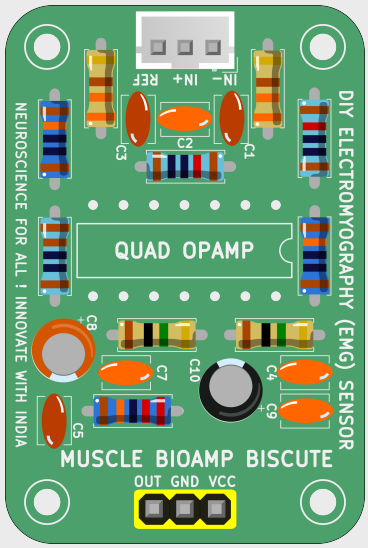


Fig. 19: Header Pins

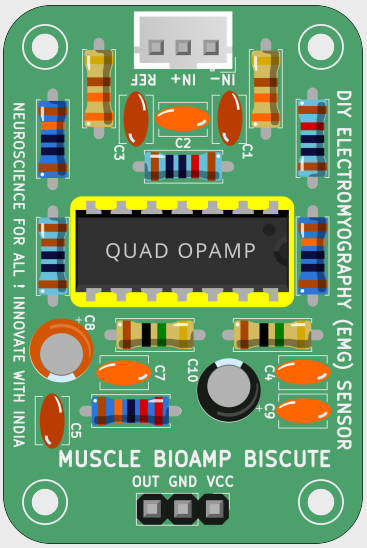


Fig. 20: IC

Still can't figure out the assembly? You can watch the video below to assemble your Biscute.

https://youtu.be/2dzW6pVT1L8

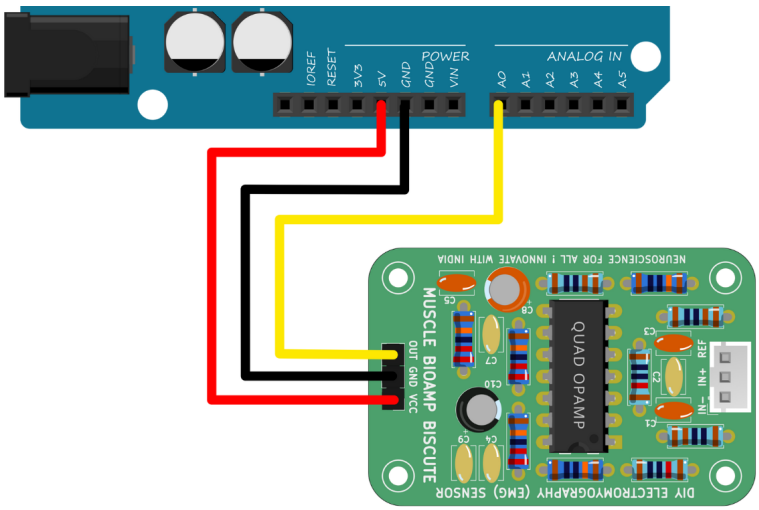
1.7 Using the kit

1.7.1 Step 1: Connect Arduino UNO R3

Connect `VCC` to either `5V` or `3.3V`, `GND` to `GND`, and `OUT` to `Analog pin A0` of your Arduino UNO via jumper cables provided by us. If you are connecting `OUT` to any other analog pin, then you will have to change the INPUT PIN in the arduino sketch accordingly.

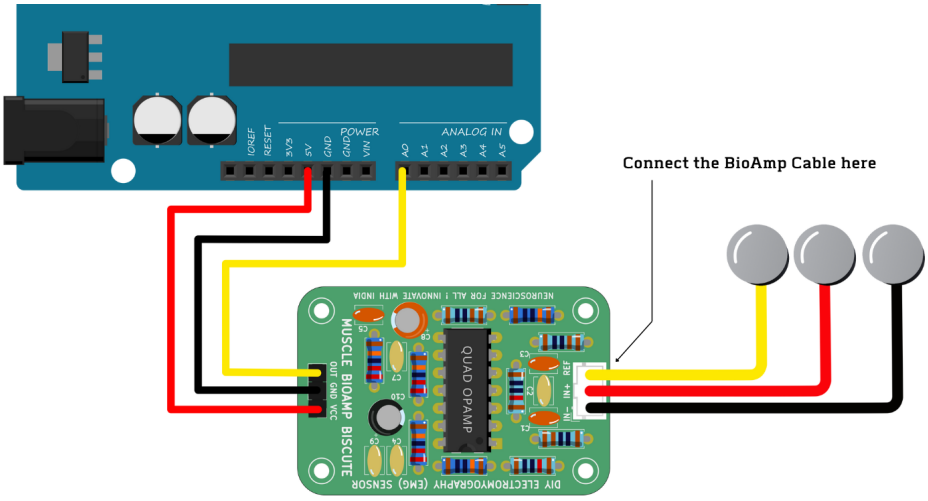
Note: For demonstration purposes we are showing connections of the sensor with Arduino UNO R3 but you can use any other development board or a standalone ADC of your choice.

Warning



Take precautions while connecting to power, if power pins (GND & VCC) are to be swapped, your sensor will be fried and it'll become unusable (DIE).

1.7.2 Step 3: Connecting electrode cable



Connect the BioAmp cable to Muscle BioAmp Biscute by inserting the cable end in the JST PH connector as shown above.

1.7.3 Step 4: Skin Preparation

Apply Nuprep Skin Preparation Gel on the skin surface where electrodes would be placed to remove dead skin cells and clean the skin from dirt. After rubbing the skin surface thoroughly, clean it with an alcohol wipe or a wet wipe.

For more information, please check out detailed step by step Skin Preparation Guide.

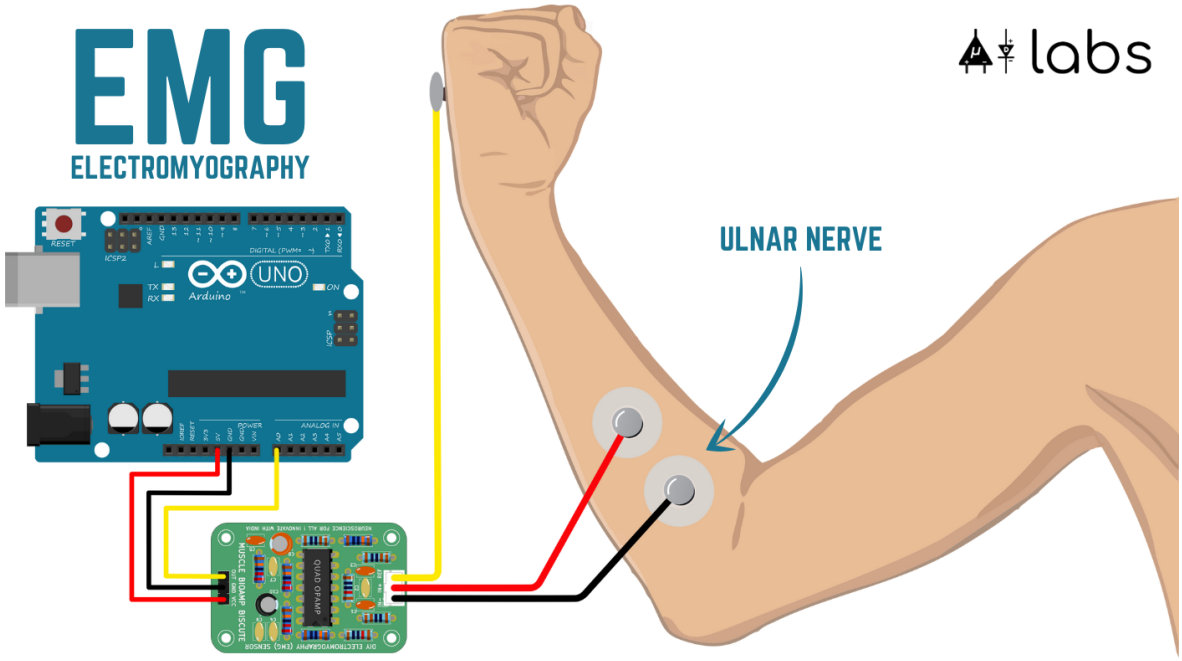
1.7.4 Step 5: Measuring EMG (ElectroMyoGraphy)

Electrodes placement

We have 2 options to measure the EMG signals, either using the gel electrodes or using dry electrode based Muscle BioAmp Band. You can try both of them one by one.

- **Using gel electrodes:**

1. Connect the BioAmp cable to gel electrodes,
2. Peel the plastic backing from electrodes
3. Place the IN+ and IN- cables on the arm near the ulnar nerve & REF (reference) at the back of your hand as shown in the connection diagram.



- **Using Muscle BioAmp Band:**

1. Connect the BioAmp cable to Muscle BioAmp Band in a way such that IN+ and IN- are placed on the arm near the ulnar nerve & REF (reference) on the far side of the band.
2. Now put a small drop of electrode gel between the skin and metallic part of BioAmp cable to get the best results.

Tip

Visit the complete documentation on how to assemble and use the BioAmp Bands or follow the youtube video given below.

Tutorial on how to use the band:

https://youtu.be/xYZdw0aesa0

Note

In this demonstration we are recording EMG signals from the ulnar nerve, but you can record EMG from other areas as well (biceps, triceps, legs, jaw etc) as per your project requirements. Just make sure to place the IN+, IN- electrodes on the targeted muscle and REF on a bony part.

Uploading the code

Connect your Arduino UNO to your laptop using the USB cable (Type A to Type B). Copy paste any one of the arduino sketches given below in Arduino IDE v1.8.19 that you downloaded earlier:

EMG Filter

EMG Envelope

Go to `tools` from the menu bar, select `board` option then select Arduino UNO. In the same menu, select the COM port on which your Arduino Uno is connected. To find out the right COM port, disconnect your board and reopen the menu. The entry that disappears should be the right COM port. Now upload the code, & open the serial plotter from the `tools` menu to visualize the EMG signals.

After opening the serial plotter make sure to select the baud rate to 115200.

Warning

Make sure your laptop is not connected to a charger and sit 5m away from any AC appliances for best signal acquisition.

Visualizing the EMG signals

For visualizing the EMG signals, use Chords Web for quick and hassle-free real-time bio-potential signals visualization right from your browser, without installing any software.

Now flex your arm to visualize the muscle signals in real time on your laptop.

Video tutorial:

https://youtu.be/ujFsAE0E0nk

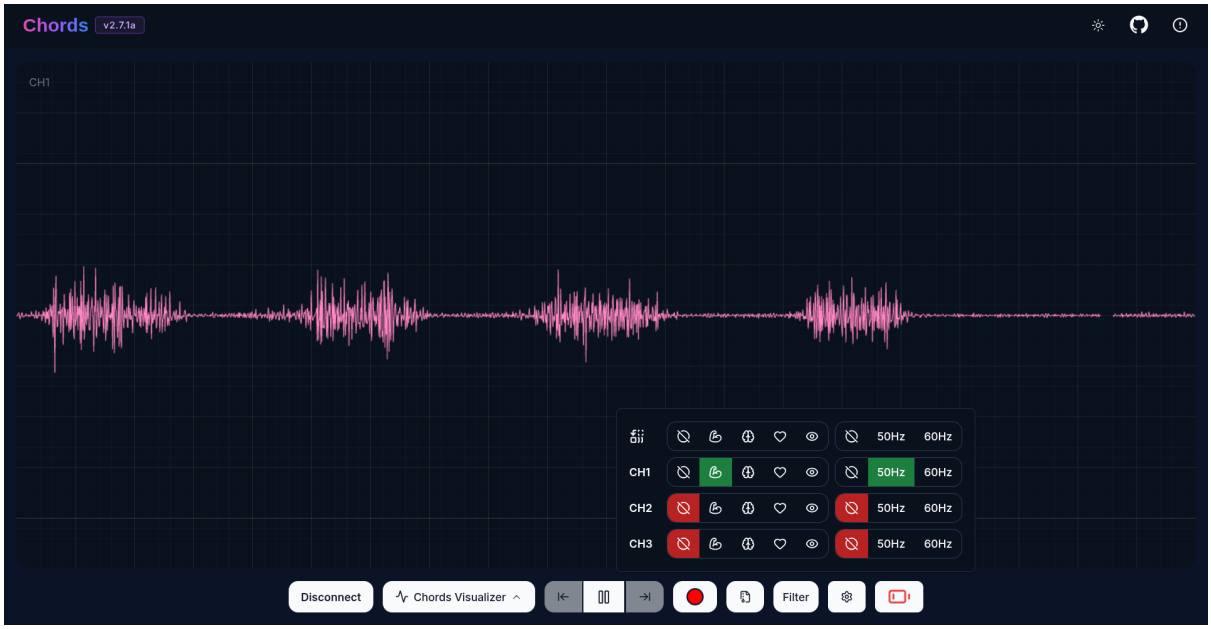


Fig. 21: Visualizing EMG signals on Chords Web

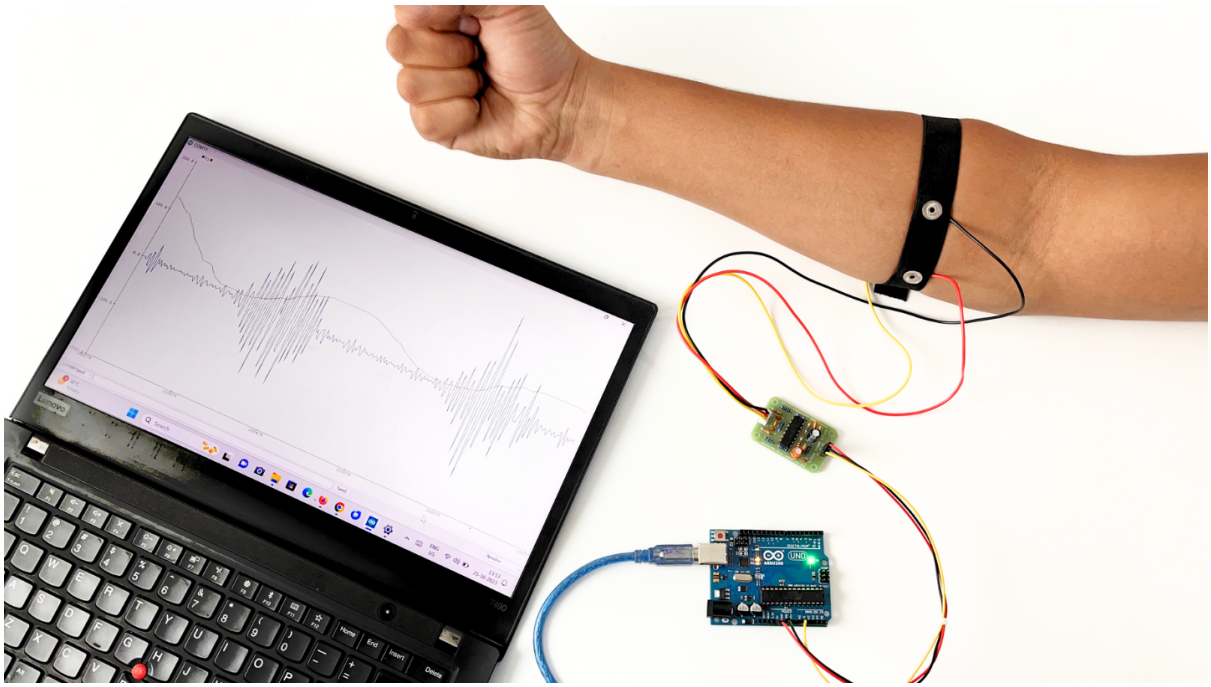


Fig. 22: Visualizing EMG signals on Arduino IDE v1.8.x